Build Compatibility -- Guided Notes

Unit 2.5 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

Name: ___________________________________ Date: ____________ Period: ______

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The most critical compatibility check in any PC build is the CPU ________. A CPU with socket LGA 1700 ________ (will/will not) fit a motherboard with socket AM4. There is ________ physical adapter or workaround that allows a CPU to be used with an incompatible socket.

Question 2

DDR4 and DDR5 RAM are physically ________ with each other because the ________ (alignment notch) is in a different position on each generation. Installing a DDR5 stick into a DDR4 slot is ________ without forcing it, which prevents accidental damage. Always verify the motherboard's ________ specification before purchasing RAM.

Question 3

GPU physical compatibility requires that the GPU ________ length fits within the case's maximum GPU clearance. A 340mm GPU ________ (will/will not) fit in a case with 310mm GPU clearance. Before buying, check the case ________ for maximum GPU length in millimeters.

Question 4

PSU wattage should cover the system's total power requirement plus a ________ % to ________ % headroom buffer. A system with a CPU TDP of 125W, GPU TDP of 200W, drives and fans totaling 50W has an estimated total TDP of ________ W. The recommended minimum PSU for this system would be ________ W.

Question 5

A PCIe x16 graphics card ________ (will/will not) fit into a PCIe x1 slot because the x16 card is physically ________ than the x1 slot. However, a PCIe x1 card ________ (will/will not) fit into an x16 slot and will operate at x1 bandwidth. PCIe slots are designed to be electrically ________ compatible.

Question 6

An ATX power supply _______ (will/will not) fit in a case designed for SFX form factor. ATX PSUs measure approximately _______ mm long, while SFX PSUs are significantly _______. When building a small form factor (SFF) system, always verify that the PSU form factor matches the _______ specifications.

Question 7

Before installing a new CPU on an existing motherboard, a technician should consult the manufacturer's _______ to verify the CPU is supported. When a new CPU generation requires a BIOS update for compatibility, the technician must update the BIOS _______ (before/after) installing the new CPU, using a/an _______ CPU that the board already supports.

Question 8 -- Real World Application

REAL WORLD: A customer brings in a PC build they assembled at home. It will not POST. They have a 13th Gen Intel Core i9 in an LGA 1700 motherboard, 2 x 16 GB DDR5-5600 RAM, and a 4 TB NVMe SSD. The motherboard is an older model manufactured before 13th Gen CPUs were released. List the compatibility checks you would perform in order and explain what you expect to find.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-5-build-compatibility/2-5-build-compatibility-answer-key.pdf
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